

Daniel Miles

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EDUCATION

- Carnegie Mellon University** | M.S. in Computer Science (Natural Language Processing) 2015 - 2017
- Thesis on contextual word embeddings for domain adaptation awarded department recognition; GPA: 3.92/4.0
- University of California, Berkeley** | B.S. in Mathematics with Computer Science Minor 2011 - 2015
- Completed 6 courses in computational linguistics and statistical NLP; summa cum laude

EXPERIENCE

- OpenAI** | Senior NLP Engineer March 2022 - Present
- Led development of retrieval-augmented generation (RAG) pipeline that improved answer relevance scores by 34% across 50M+ daily queries in production systems
 - Architected and shipped sequence classification module serving 2.8B monthly inference requests with 99.98% uptime, mentoring team of 4 junior engineers on best practices
 - Optimized transformer tokenization layer reducing inference latency by 47% and cutting infrastructure costs by \$1.2M annually
- Anthropic** | Senior Machine Learning Engineer - NLP July 2020 - February 2022
- Designed and deployed constitutional AI fine-tuning framework reducing harmful outputs by 72% while maintaining performance across 15+ benchmark datasets
 - Established open-source NLP library with 8.2K GitHub stars, enabling 400+ researchers to build safer language models
 - Led evaluation infrastructure project processing 500K+ model generations daily, cutting evaluation time by 63% through novel sampling strategies
- Scale AI** | Lead NLP Engineer January 2019 - June 2020
- Pioneered named entity recognition system for financial documents achieving 96.2% F1 score across 12 entity types, processing 8M+ documents annually
 - Managed cross-functional team of 3 engineers delivering semantic search infrastructure processing 15M queries/month with <200ms p99 latency
 - Contributed foundational work to open-source spaCy project with 5 merged pull requests improving transformer-based NER by 8%
- Databricks** | NLP Engineer August 2017 - December 2018
- Developed multilingual sentiment analysis framework supporting 23 languages, reducing annotation costs by 58% through active learning strategies
 - Optimized BERT inference on Spark clusters achieving 12x throughput improvement, enabling real-time processing for 500+ enterprise customers
 - Implemented distributed question-answering pipeline on 100M document corpus with 89% accuracy, deployed to 3 Fortune 500 companies

PROJECTS

- Domain-Adaptive Prompt Tuning Framework** | Python, PyTorch, Hugging Face Transformers, PEFT 2021 - 2022
- Created parameter-efficient fine-tuning methodology achieving 91% of full-model performance with 99.8% fewer trainable parameters on specialized NLP tasks
 - Published methodology in 3 peer-reviewed venues with 240+ citations, demonstrating practical applications across biomedical, legal, and clinical domains
- Zero-Shot Cross-Lingual Transfer System** | Multilingual BERT, mBERT, XLM-R, Kubernetes 2019 - 2020
- Engineered zero-shot transfer learning system enabling 7 low-resource languages to leverage English training data with only 4% performance degradation
 - Deployed containerized service handling 2M+ monthly inference requests with automatic language detection, reducing manual labeling costs by \$340K annually

EXTRACURRICULAR

- ACL (Association for Computational Linguistics)** | Reviewer & Workshop Organizer 2019 - Present
- Reviewed 40+ papers annually for top-tier NLP conferences (ACL, EMNLP, NAACL), maintaining consistent acceptance standards
 - Organized 2 workshops on parameter-efficient fine-tuning attracting 300+ attendees from academia and industry
- Hugging Face Community** | Contributor & Ambassador 2020 - Present
- Contributed 12 model implementations to Transformers library, accumulating 8.5K+ model downloads monthly

- Maintained 2 open-source NLP toolkits with 3.2K GitHub stars and mentored 15+ new open-source contributors

TECHNICAL SKILLS

Languages – Python (expert), SQL (advanced), Bash (intermediate), Java (intermediate)

Technologies – Transformers (BERT, GPT, T5, Llama), PyTorch, TensorFlow, Hugging Face, spaCy, NLTK, Keras, LangChain, LLaMA Index, RAG systems, MLflow, Weights & Biases

Tools – Python, SQL, Git, Docker, Kubernetes, Apache Spark, Ray, AWS (SageMaker, EC2), Weights & Biases, Airflow, Jupyter, VSCode